

Report 1: Neuro-Symbolic Reasoning in the Age of Large Reasoning Models: A Path Toward NSRMs.

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Abstract

Only within the past few years have neurosymbolic methods been introduced to LLMs. Neurosymbolic AI aimed to improve symbolic reasoning and achieve more faithful results in LLMs. However, recently, large reasoning models (LRMs) have grown in popularity to solve this problem in LLMs. Even though LRMs are not as faithful as neurosymbolic pipelines because they lack formal reasoning, they are performing well on popular symbolic datasets such as FOLIO. This report will go over why neurosymbolic methods remain superior to LRMs through evaluating reasoning models and comparing them against a top-tier neurosymbolic method (LTRAG), which outperforms LINC and Logic-LM on FOLIO. This report finds that using LTRAG offers a consistent, cost-efficient, faithful accuracy boost to results compared to reasoning models. For example, using a cheap inference model like GPT-4o mini achieves a faithful 71.98% accuracy and costs \$0.15. Meanwhile, the newer reasoning model, GPT 5.4 mini, on medium effort using CoT achieves a 65.93% accuracy and costs \$0.81. We also discuss the future of neurosymbolic AI with NSRMs (Neuro-Symbolic Reasoning Models), a path where LRMs can work with neurosymbolic pipelines to achieve faithful and efficient reasoning.

1 Introduction

Large language models are growing more and more powerful by the day. At the rate they are growing at, it is becoming increasingly more difficult to outperform pure neural methods on reasoning benchmarks. However, neurosymbolic methods offer consistency in accuracy through reduced hallucinations compared to pure neural methods. One fundamental reasoning task that LLMs seem to struggle on is logical reasoning. This is why researchers came up with methods to stress test LLMs on reasoning tasks such as ProofWriter, a synthetic benchmark for rule-based reasoning (Tafjord et al., 2021);

FOLIO, a first-order logic (FOL) based benchmark (Han et al., 2024); and AR-LSAT, a benchmark sampled from the LSAT exam (Zhong et al., 2021). These logic datasets are listed in order of difficulty for LLMs. We will be focusing on FOLIO, which involves human-annotated FOL examples. FOLIO provides FOL premises and a conclusion derived from those premises. The conclusion can either be true, false, or uncertain.

To solve such benchmarks, neural researchers resorted to improving neural methods. Neural methods, such as CoT prompting, improved the capability of LLMs for solving logic and symbolic reasoning tasks (Wei et al., 2023). This eventually led to the development of large reasoning models (LRMs). LRMs are trained to master CoT through RL feedback loops and are trained on reasoning data. Furthermore, "think" extensively on inference time (Xu et al.), which is adjusted by "reasoning effort." While these neural techniques prove to be effective at improving results, their weakness revolves around the faithfulness of their reasoning because there is no evidence citing that they actually think about inference or how they reason over their output. (Wei et al., 2023).

Other researchers with papers such as Logic-LM (Pan et al., 2023) aimed to use neurosymbolic methods to solve logical reasoning problems, which also solves faithfulness in reasoning problems in neural reasoning by using sound theorem provers such as Prover9 or Z3 to produce fortified reasoning with a robust inference trace and only relying on the LLM for translating to the solvers, aka semantic parsing. Logic-LM uses self-refinement to produce accurate semantic parsing. Other papers like LINC (Olausson et al., 2023) use k-maj voting by having a theorem prover solve k LLM FOL semantic parses and picking the majority solution. LINC managed to achieve outstanding results on ProofWriter and dominated that benchmark. However, it struggled on FOLIO against neural methods. More recently,

LTRAG (Hu et al., 2025) uses more advanced self-refinement than Logic-LM through agentic RAG pipelines to accurately semantically parse FOLIO and AR-LSAT examples, producing better results than LINC and Logic-LM. Also, advanced concepts that emerged from these papers are using CoT for semantic parsing, which in turn gets the best of both worlds.

This report compares LTRAG to neural methods, such as reasoning models, and CoT on the FOLIO benchmark to analyze the accuracy and efficiency of these approaches on FOLIO. Furthermore, we evaluate the faithfulness of their accuracy based on its consistency and output choices. For example, a method is more faithful if it achieves improved accuracy more consistently and evaluates to true, false, or uncertain on the FOLIO conclusion more confidently. Efficiency is evaluated through cost-effectiveness. However, in the future, we will discuss more measurements, such as environmental factors. This report also dives into future directions in the field and how LRMs can work with neurosymbolic AI to create a potential NSRM (Neuro-Symbolic Reasoning Model).

2 Related Work

A similar study has been performed by McGinness and Baumgartner (McGinness and Baumgartner, 2025). They evaluate newer reasoning models and compare them to inference models + a semantic solver (Z3) to solve a reasoning benchmark, ProntoQA, which was also tested in Logic-LM. They find that the smaller inference model (Gemma 3 12B) + Z3 achieves a similar accuracy to DeepSeek R1 at a fraction of the cost. This study is extremely insightful and helped prove one of my initial hypotheses.

My initial hypothesis was that neurosymbolic methods would outperform reasoning models on reasoning tasks. However, one drawback of their findings is that they only test this on ProntoQA, a dataset that Logic-LM is clearly good at. I aim not to put their research down but to expand on it by testing more datasets, more neurosymbolic methods, and newer reasoning models and quantifying reasoning at inference time to reasoning effort levels, which OpenAI makes possible. However, at the time of writing this report, I am only doing my experiment on FOLIO with a variety of models.

I agree with their reasoning behind neurosymbolic methods being more effective. However, I

aim to find out to what extent. Especially now that agentic AI is approaching, neurosymbolic methods are more generalizable than ever, and tool calling a theorem prover is something modern LLMs are capable of.

3 Framework

For this experiment, we used LTRAG's (Hu et al., 2025) pipeline and their definition of "standard" and "CoT" to test various models. The LTRAG pipeline relies on using a "train" FOLIO dataset for RAG to help the LLM produce semantically accurate FOL. It also uses COT. If the parsing fails, then the LLM retries this time with RAG from a "fixer" knowledge base to retry parsing.

Through LTRAG's codebase, I found "standard" to follow more of a CoT-style prompt. Specifically because of the prompt: "You need to emphasize which premises were used to obtain your results." LTRAG defines CoT as a two-shot example of a COT-style response fed to the LLM to see how it is supposed to translate the knowledge base accurately.

For Standard and CoT there is a parser LLM that simply pulls the answer from the LLM's response. However, for LTRAG, the parser pulls the premises and conclusion and sends them to Z3.

Because LTRAG requires the parser to retrieve a solid response through Z3, we decided to include the parse in the cost of LTRAG's run. This is why LTRAG tends to be more expensive than its counterparts. However, even then, using it with efficient models can be cheaper than using reasoning models.

For cost analysis, we consider input, output, and reasoning tokens for each model used. The cost per million token is derived from OpenAI's [web-site](#). Because OpenAI offers this transparency, we decided to use OpenAI's models for this report.

Our model selection revolves around aiming to understand how inference models and reasoning models perform on COT, standard, and LTRAG on FOLIO. The primary model used to compare our results to LTRAG's is GPT-3.5 Turbo. Furthermore, we decided to add on two more newer inference models: GPT 4.1 mini and GPT 4o mini. As for reasoning models, we add GPT 5.4 nano and GPT 5.4 mini. We specifically chose those models because they are the newest OpenAI reasoning models that are small and allow us to control and measure their reasoning effort at inference time,

defined by none, low, medium, high, and Xhigh. However, because of expenses, we decide to limit the amount of reasoning efforts we test.

4 Results

For our results we test our framework and output the results in our tables on the following pages. As for our graph, we test our initial hypothesis on a modified LINC to see if we can attain something similar to the researchers in our related works section.

5 Discussion

5.1 Initial Hypothesis

Figure 1 shows us that our initial hypothesis is valid. myLINC uses LINC’s pipeline but modifies it to include no k-maj voting. Also, we ran the experiment on only 70 samples of each FOLIO and ProofWriter. The goal of this wasn’t to produce something publishable but rather to confirm what the researchers in the related works did on a different pipeline and get proof for our initial hypothesis. This test accurately shows that our inference model (GPT 4.1 mini) outperforms the neural baseline of a comparable reasoning model (GPT 5.4 nano) on both ProofWriter and FOLIO. They are comparable because OpenAI’s [website](#) gives us a way to determine that GPT 4.1’s intelligence and speed are equal to GPT 5.4 nano’s reasoning and speed. We theorize that the reason that GPT 5.4 nano doesn’t match with standard or medium in Table 2 is because LTRAG has an idea of "standard" that involves a degree of CoT reasoning that inflates the results. Also, the graph only tests on 70 FOLIO samples. The next time we run our experiments, we shall redefine the idea of standard and CoT in the LTRAG pipeline to match the standard idea of those two principles. The few-shots on CoT aren’t as misleading. However, the "standard" prompt can be misleading. We theorize that this is the reason standard sometimes outperforms CoT in our tables.

Table 1 further proves that we did match LTRAG’s pipeline on standard, CoT, and LTRAG because our results match theirs on GPT 3.5 Turbo. There is a small variance, but it is not significant. (Hu et al., 2025). This means to me that LTRAG did intend "standard" to be as it exists in their codebase, with some CoT element to it. However, we do not agree with that design as much and might change it for future tests to produce more accurate raw results based on a more common understanding

of "standard," and we shall call CoT few-shot CoT or remove few shots entirely. However, for this report and all our results, we adhere to LTRAG’s idea of these methods to remain consistent.

5.2 Inference Models

For the sake of this report, we are distinguishing reasoning-based LLMs from classic LLMs by referring to classic LLMs as inference models and reasoning-based LLMs as reasoning models and LRMs. In Table 1 we carry out tests on OpenAI’s newer inference models, such as GPT 4.1 mini and GPT 4o mini. As we can see, LTRAG outperforms the other methods on GPT-4o mini and at a small cost compared to the other inference models. However, on a newer GPT-4.1 mini, standard and CoT outperform LTRAG. We theorize a possibility where 4.1 mini could have been trained on FOLIO. Therefore, we must add more datasets to solidify our results. However, as for now, this model is highly competitive and performs rather well on logical reasoning. The distinguishing factor between neurosymbolic methods and neural methods is that neurosymbolic methods are backed by sound theorem provers. Therefore, we are able to achieve faithful accuracy scores. To test the faithfulness of neural models and methods, we carried out the same experiment for reasoning models across varying reasoning efforts in Tables 2 and 3.

5.3 Reasoning Models

The reasoning models we tested seem to offer less faithful results and are inefficient (costly). The lack of faithfulness comes from the lack of consistency we can see across GPT 5.4 mini in Table 2 and GPT 5.4 Nano in Table 3. Nano seems to outperform mini on almost every mode. However, diving deeper into the results, we can see that mini tends to pick "uncertain" much more frequently than nano. We are not sure how to quantify this measurement yet. However, we will come up with a more solid measurement within our next report. For now, just by analyzing their responses, we were able to detect that mini was being more defensive than nano. Therefore, mini was more conservative of the answers it chose.

Our analysis based on their choices doesn’t tell us that mini is a worse model than nano. Rather, mini is more faithful. Unironically, this dilemma is what neurosymbolic researchers resolved years ago. While LTRAG doesn’t perform the best, it is most faithful because it uses sound theorem provers for

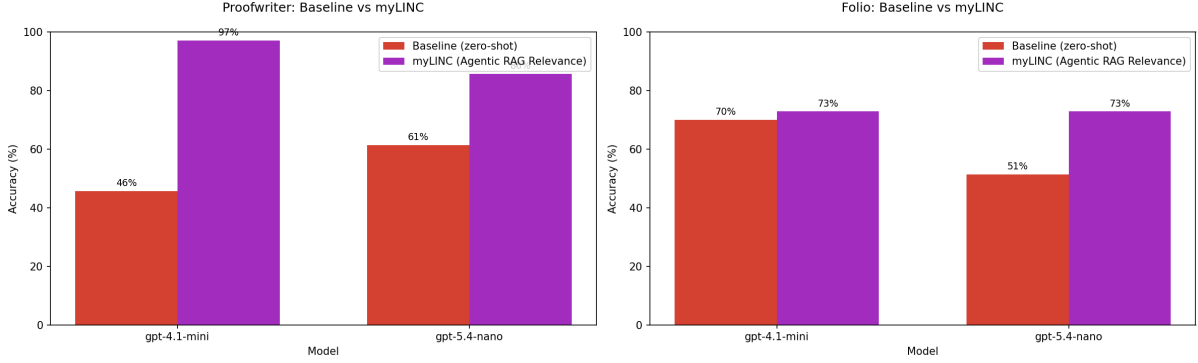


Figure 1: Accuracy comparison of Baseline vs myLINC on ProofWriter and FOLIO datasets across GPT-4.1-mini and GPT-5.4-nano.

Model	FOLIO (Accuracy %)			Cost (\$)		
	LTRAG	Standard	CoT	LTRAG	Standard	CoT
GPT-4o-mini	71.98	69.23	67.03	0.15	0.07	0.12
GPT-4.1-mini	76.92	81.87	81.87	0.38	0.32	0.35
GPT-3.5-turbo	69.78	54.40	62.09	0.53	0.12	0.31

Table 1: Non-reasoning baselines on FOLIO (182 samples). Accuracy (%) and cost (\$) per pipeline. Bold marks the highest accuracy and the lowest cost per row.

reasoning. The only difference is neurosymbolic methods are able to get consistent faithful results at a fraction of the cost of a reasoning model. This is the reason we highlight GPT-4o mini achieving 71.98% accuracy at \$0.15 from Table 1, while GPT 5.4 mini achieves 65.93% accuracy at \$0.81 in Table 3. We highlight this example not to show that neural methods are inherently worse. Rather, to show that they can be less faithful and less efficient than neurosymbolic methods.

6 Conclusion

Neurosymbolic methods tend to be more consistent and more efficient at solving the FOLIO logic benchmark than neural methods. Neurosymbolic methods are highly effective and consistent and tend to perform well even on smaller models.

Our conclusion doesn’t mean we should completely ditch neural methods. After all, neural methods allow for proper reasoning over premises to construct structured FOL representations that a symbolic solver can solve. LLMs are largely powerful. Neurosymbolic methods suggest a different way to use them. A way that is more effective using intricate neurosymbolic pipelines.

Reasoning models can even be a great tool for semantic parsing. Ironically reasoning models might be the reason neurosymbolic methods get adopted more. This is because reasoning models tend to be

great at tool use. A great model should be able to interact with a theorem prover and produce sound results like never before. We believe in the strength of reasoning models alongside neurosymbolic methods. However, before then, research like this report is necessary to highlight the benefit of neurosymbolic methods.

Future work might include developing tool calling agents that can effectively use popular neurosymbolic methods such as the ones discussed in this paper. We strongly believe in a future where NSRMs (Neuro-Symbolic Reasoning Models) exist. A model that chooses faithful symbolic reasoning by tool-calling neurosymbolic pipelines such as LTRAG.

Limitations

We need consistent tests in order to better defend our hypothesis. Based on our current results, we are seeing patterns that show neurosymbolic methods tend to be more faithful and efficient than neural methods. However, our data is limited so far. In order to better generalize, we have to introduce more neurosymbolic methods to our framework and add more datasets. We are currently working on adding the AR-LSAT benchmark next to our results.

Another limitation is that we have yet to quantify faithfulness in a measurable way. As of now,

Effort	FOLIO (Accuracy %)			Cost (\$)		
	LTRAG	Standard	CoT	LTRAG	Standard	CoT
None	75.27	59.89	73.08	0.26	0.15	0.17
Low	75.27	78.02	74.18	0.35	0.19	0.20
Medium	81.32	81.87	79.12	0.48	0.27	0.27
High	79.67	81.87	80.22	0.59	0.32	0.31

Table 2: GPT-5.4-nano on FOLIO across reasoning-effort levels. Accuracy (%) and cost (\$) per pipeline. Bold marks the highest accuracy and the lowest cost per row.

Effort	FOLIO (Accuracy %)			Cost (\$)		
	LTRAG	Standard	CoT	LTRAG	Standard	CoT
None	72.53	64.84	60.44	0.85	0.38	0.52
Low	75.82	67.03	64.29	0.97	0.48	0.58
Medium	75.82	75.82	65.93	1.82	0.73	0.81

Table 3: GPT-5.4-mini on FOLIO across reasoning-effort levels. Accuracy (%) and cost (\$) per pipeline. Bold marks the highest accuracy and the lowest cost per row.

faithfulness is backed largely by observing that neurosymbolic methods tend to have a higher true positive rate and a true negative rate than neural methods. Also, the larger GPT 5.4 tends to be more defensive, similar to neurosymbolic methods, and thus more faithful. However, we have yet to come up with a way to quantify that as well. Both of these limitations are things we shall address in our next report to better defend our hypothesis.

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References

- Leonardo de Moura and Nikolaj Bjørner. 2008. [Z3: An Efficient SMT Solver](#). In *Tools and Algorithms for the Construction and Analysis of Systems*, pages 337–340, Berlin, Heidelberg. Springer.
- Simeng Han, Hailey Schoelkopf, Yilun Zhao, Zhen-ting Qi, Martin Riddell, Wenfei Zhou, James Coady, David Peng, Yujie Qiao, Luke Benson, Lucy Sun, Alex Wardle-Solano, Hannah Szabo, Ekaterina Zubova, Matthew Burtell, Jonathan Fan, Yixin Liu, Brian Wong, Malcolm Sailor, and 16 others. 2024. [FOLIO: Natural Language Reasoning with First-Order Logic](#). *arXiv preprint*. ArXiv:2209.00840 [cs.CL].
- Ruikang Hu, Shaoyu Lin, Yeliang Xiu, and Yongmei Liu. 2025. [LTRAG: Enhancing Autoformalization and Self-refinement for Logical Reasoning with Thought-Guided RAG](#). In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 2483–2493, Vienna, Austria. Association for Computational Linguistics.
- Nidhal Jegham, Marwan Abdelatti, Chan Young Koh, Lassad Elmoubarki, and Abdeltawab Hendawi. 2025. [How Hungry is AI? Benchmarking Energy, Water, and Carbon Footprint of LLM Inference](#).
- Lachlan McGinness and Peter Baumgartner. 2025. [Large Language Models Imitate Logical Reasoning, but at what Cost?](#) *arXiv preprint*. ArXiv:2509.12645 [cs.AI].
- Theo X. Olausson, Alex Gu, Benjamin Lipkin, Cedegao E. Zhang, Armando Solar-Lezama, Joshua B. Tenenbaum, and Roger Levy. 2023. [LINC: A Neurosymbolic Approach for Logical Reasoning by Combining Language Models with First-Order Logic Provers](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 5153–5176. ArXiv:2310.15164 [cs.CL].
- Liangming Pan, Alon Albalak, Xinyi Wang, and William Wang. 2023. [Logic-LM: Empowering Large Language Models with Symbolic Solvers for Faithful Logical Reasoning](#). In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 3806–3824, Singapore. Association for Computational Linguistics.
- Oyvind Tafjord, Bhavana Dalvi, and Peter Clark. 2021. [ProofWriter: Generating Implications, Proofs, and Abductive Statements over Natural Language](#). In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, pages 3621–3634, Online. Association for Computational Linguistics.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. 2023. [Chain-of-Thought Prompting](#)

Elicits Reasoning in Large Language Models. *arXiv preprint*. ArXiv:2201.11903 [cs.CL].

Fengli Xu, Qianye Hao, Zefang Zong, Jingwei Wang, Yunke Zhang, Jingyi Wang, Xiaochong Lan, Jiahui Gong, Tianjian Ouyang, Fanjin Meng, Chenyang Shao, Yuwei Yan, Qinglong Yang, Yiwen Song, Si-jian Ren, Xinyuan Hu, Yu Li, Jie Feng, Chen Gao, and Yong Li. Towards Large Reasoning Models: A Survey of Reinforced Reasoning with Large Language Models.

Jundong Xu, Hao Fei, Liangming Pan, Qian Liu, Mong-Li Lee, and Wynne Hsu. 2024. [Faithful Logical Reasoning via Symbolic Chain-of-Thought](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13326–13365, Bangkok, Thailand. Association for Computational Linguistics.

Wanjun Zhong, Siyuan Wang, Duyu Tang, Zenan Xu, Daya Guo, Jiahai Wang, Jian Yin, Ming Zhou, and Nan Duan. 2021. [AR-LSAT: Investigating Analytical Reasoning of Text](#). *arXiv preprint*. ArXiv:2104.06598 [cs.CL].